



## \* Practical - 5 \*

\* Handler and Threads Develop an application that ~~provides facility to set~~ should increment the value of a text after specific interval of time. Also provide feature to stop incrementing value of the text.

\* activity\_main.xml

<LinearLayout

xmlns:android="http://schemas.android.com/apk/res/android"

android:layout\_width="match\_parent"

android:layout\_height="match\_parent"

android:orientation="vertical"

android:gravity="center"

android:padding="16dp">

<TextView

android:id="@+id/textViewCounter"

android:layout\_width="wrap\_content"

android:layout\_height="wrap\_content"

android:text="0"

android:textSize="48sp"

android:layout\_marginBottom="24dp"

android:textStyle="bold"/>



<Button

```
android:id="@+id/btnStart"  
android:layout_width="wrap_content"  
android:layout_height="wrap_content"  
android:text="Start"  
android:layout_marginBottom="16dp"/>
```

<Button

```
android:id="@+id/btnStop"  
android:layout_width="wrap_content"  
android:layout_height="wrap_content"  
android:text="Stop"/>
```

</LinearLayout>

\* MainActivity.java

```
package com.example.myapplication;
```

```
import android.os.Bundle;
```

```
import android.os.Handler;
```

```
import android.widget.Button;
```

```
import android.widget.TextView;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```
public class MainActivity extends AppCompatActivity {
```





```
private TextView textViewCounter;  
private Button btnStart, btnStop;
```

```
private int counter = 0;  
private boolean isRunning = false;  
private Handler handler = new Handler();  
private Runnable runnable;
```

⑨ Override

```
protected void onCreate (Bundle savedInstanceState)  
super.onCreate (savedInstanceState);  
setContentView(R.layout.activity_main);
```

```
textViewCounter = findViewById (R.id.textViewCounter);  
btnStart = findViewById (R.id.btnStart);  
btnStop = findViewById (R.id.btnStop);
```

```
runnable = new Runnable () {
```

⑨ Override

```
public void run() {  
    if (isRunning) {  
        counter++;  
        textViewCounter.setText (String.  
            valueOf (counter));  
        handler.postDelayed (this, 1000);  
    }  
}
```

```
};
```



```
btnStart.setOnClickListener(v -> {  
    if (C !isRunning) {  
        isRunning = true;  
        handler.post(Crunnable);  
    }  
});
```

```
btnStop.setOnClickListener(v -> {  
    isRunning = false;  
    handler.removeCallbacks(Crunnable);  
});  
}
```

\* Output \*

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Start

Stop

Shubh